

SUMMARY TABLE: CORE MATHEMATICS GRADE 10

Sub-Strand 1.1: Real Numbers — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 1.1: Real Numbers
Driving Question	DRIVING QUESTION: Why can we never write some everyday numbers down completely — and how do mathematicians and Kenyan professionals work accurately with numbers that go on forever?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 What Kind of Number Is That? (Predict & Classify)	A cylindrical aluminium sufuria with a diameter of 28 cm from Gikomba Market; the calculator result of $28 \times \pi = 87.96459430\dots$ with digits that never end or repeat; the contrast between $28 \times 3 = 84$ (exact) and $28 \times \pi = 87.96459430\dots$ (inexact); a visual scroll of the digits of π continuing without pattern.	Some numbers — like 60.50, 160, 34.5 — can be written down completely and exactly, while others — like π and $\sqrt{2}$ — cannot be written as terminating or repeating decimals; the circumference of a circle with a whole-number diameter may not itself be a 'nice' number; mathematicians classify numbers into different types based on whether they can be expressed exactly.	We cannot yet fully explain why multiplying a whole number (28) by π produces a number that goes on forever without repeating, or what mathematical property separates numbers like π from numbers like $1/3$ or 60.50; we cannot yet explain whether $22/7$ is the same as π or merely close to it; we cannot yet name the two categories of numbers we have identified, or describe the full system of real numbers.
Lesson 2 Sorting the Number World: Classifying Real Numbers from Kenyan Contexts	Learners sorted 20 Kenyan number cards — including prices in Kenyan shillings, distances in km, temperatures in $^{\circ}\text{C}$, fractions, repeating decimals, and irrational numbers such as π and $\sqrt{2}$ — into self-invented groups, then re-sorted them according to the formal real number classification system using a projected nested Venn diagram.	The real number system is organised as nested sets: natural numbers $\subset$ integers $\subset$ rational numbers $\subset$ real numbers, with irrational numbers occupying the region inside $\mathbb{R}$ but outside $\mathbb{Q}$. Rational numbers can always be written as p/q and produce terminating or repeating decimals; irrational numbers — such as π and $\sqrt{2}$ — cannot be written as p/q and produce non-terminating, non-repeating decimals.	π is an irrational number. Because it cannot be expressed as a fraction p/q , the circumference of the mama mboga's sufuria (28π cm) cannot be written as an exact decimal — the digits 87.96459430... continue forever without repeating. This is why the daughter's string always has a tiny gap or overlap no matter how many decimal places she uses.
Lesson 3 Lesson 3 — "Rational or Irrational? Sorting the Number Family"	Learners observed the decimal expansions of ten Kenyan everyday numbers on a calculator and discovered that some decimals terminate (60.5), some repeat in a	Learners learned that rational numbers are expressible as p/q (integers, $q \neq 0$), that terminating and recurring decimals are always rational, that irrational numbers	The formal classification of real numbers into rational and irrational subsets, the algebraic technique for converting a recurring decimal to a fraction, and the

	pattern ($0.142857142857... = 1/7$; $0.333... = 1/3$; $22/7 = 3.142857...$), and others — particularly π and $\sqrt{2}$ — neither terminate nor repeat across all visible digits.	have non-terminating non-repeating decimal expansions, and that π is irrational — which is exactly why multiplying the sufuria's diameter (28 cm) by π produces a circumference that can never be written as an exact fraction or a terminating decimal.	direct connection between π's irrationality and the unavoidable gap or overlap when cutting a string to fit the sufuria rim were explained and practised.
Lesson 4 Placing Numbers on the Line: Can We Always Find the Exact Spot?	Learners constructed a classroom string number line (–5 to 5) and pegged 8 number cards (including $-\sqrt{3}$, $\sqrt{2}$, $\pi - 3$, $2.333...$) to their correct positions after computing decimal approximations. They also plotted and classified 7 Kericho daily temperatures as rational or irrational on a mini number-line strip.	Every real number — whether rational or irrational — occupies a unique, precise point on the number line. Irrational numbers like $\sqrt{2}$ and $\pi - 3$ cannot be written exactly as fractions or terminating decimals, but they can be located to any desired accuracy using decimal approximations. Non-terminating repeating decimals (e.g. $2.333...$) are rational. The daughter's 88 cm string is a rational approximation pegged near — but never exactly at — the irrational circumference $\pi \times 28$.	The real number line is complete and continuous: between any two rational numbers there are infinitely many irrational numbers, and every irrational number has a definite address on the line even though we can never write that address in full. This is why the ugali-pot daughter's string always has a tiny gap — she is placing a rational peg at an irrational address, and no matter how many decimal places her calculator shows, she can only get closer, never exact.
Lesson 5 Operating on Real Numbers — Can We Calculate Exactly with Numbers That Never End?	Learners computed the results of adding, subtracting, multiplying, and dividing pairs of real numbers drawn from Kenyan marketplace and construction contexts (unga prices, the sufuria circumference, pharmacist doses in Kericho, jua kali fabrication in Kamukunji). They observed that most combinations follow predictable rules — but that irrational $\times$ irrational can sometimes yield a rational result ($\sqrt{2} \times \sqrt{2} = 2$), and that replacing π with $22/7$ makes the sufuria calculation rational but introduces an approximation error.	The four operations on real numbers obey BODMAS. When two rationals are combined by any operation (result defined), the result is always rational. When a rational is combined with an irrational by addition or subtraction, the result is always irrational. When a rational is multiplied or divided by an irrational, the result is generally irrational. When two irrationals are combined, the result can be either rational or irrational — there is no single rule. Keeping the exact symbolic form (e.g. 28π) is more precise than any decimal approximation.	The daughter's calculator shows $87.96459430...$ because $28 \times \pi$ is the product of a rational (28) and an irrational (π), making the result irrational — a number that cannot be written as a terminating or repeating decimal, no matter how many digits the calculator displays. When she rounds to 88 cm or uses $22/7$ instead of π, she converts the answer into a rational number — gaining the ability to write it down exactly — but at the cost of introducing a small error. This is why there is always a tiny gap or overlap in her string: exactness and writability are fundamentally in tension for irrational numbers.
Lesson 6 Operating on Real Numbers: BODMAS in the Market and the Sufuria	Learners evaluated multi-step BODMAS expressions using real numbers drawn from Gikomba Market scenarios: total market bills involving mixed numbers and decimals (unga, sukuma wiki, tomatoes), string-cost calculations using $\pi \approx 3.142$, and a composite expression involving indices and surds (4^2, $\sqrt{16}$) applied to nyama choma pricing.	The order of operations (BODMAS) governs the evaluation of expressions involving ALL real numbers — both rational (prices, weights, distances) and irrational (π, $\sqrt{2}$). Rounding an irrational number before completing all operations introduces avoidable approximation error; the most accurate strategy is to carry the irrational value symbolically (or to maximum decimal places) and round only at the final step.	When the daughter in Gikomba multiplies $28 \times \pi$ and rounds π to 3.142 first, she applies BODMAS correctly but introduces a small rounding error because π is irrational and 3.142 is only an approximation. BODMAS does not change based on the type of real number — the same rules that govern Ksh 60.50×2.5 also govern $\pi \times 28$. Professional accuracy (e.g., a civil engineer sizing a water-tank rim in Kisumu) requires carrying irrational values to as many decimal places as the context demands and

			rounding only at the final answer.
Lesson 7 Absolute Value in Action — How Far From Zero on the Number Line?	Learners observed that evaluating $ 88 - 28\pi $ using the piecewise definition of absolute value gives approximately 0.035 cm — a small but non-zero gap — and that this error is itself an irrational number. They also observed that absolute value always produces a non-negative result for any real number, whether integer, fraction, or irrational expression such as $ \pi - 3 $ or $ \sqrt{2} $.	Absolute value $ x $ measures the distance of a real number x from zero on the number line and is defined as: $ x = x$ when $x \geq 0$, and $ x = -x$ when $x < 0$. The absolute difference $ a - b $ measures the error between any two real numbers. The daughter's string error is $ 88 - 28\pi = 88 - 28\pi \approx 0.035$ cm — rational minus irrational, which is irrational — meaning the error can never be expressed as an exact fraction either.	The Ugali Pot Problem now has a precise, measurable answer: the daughter's 88 cm string is approximately 0.035 cm longer than the exact circumference 28π cm. This tiny overlap is itself an irrational number — it goes on forever without repeating. No matter how many decimal places of π she uses, the string error will always be a non-terminating, non-repeating number. Absolute value gives mathematicians and Kenyan professionals (engineers, traders, pharmacists) a rigorous, always-positive way to state 'how wrong' any rational approximation of an irrational measurement really is.
Lesson 8 Absolute Value: Distance from Zero — and the Final Explanation of the Real Number System	Learners plotted Kericho weekly temperature values (including negatives such as -3.5°C) on a number line, drew arrows from zero to each value, and measured arrow lengths. They also computed absolute values of Gikomba Market daily profit/loss figures and calculated $ 28\pi - 88 \approx 0.035$ cm as the Ugali Pot cutting error.	Absolute value $ x $ is the distance of a real number x from zero on the number line. It is always non-negative. Formally: $ x = x$ if $x \geq 0$, and $ x = -x$ if $x < 0$. Absolute value applies equally to rational numbers (e.g., $ -Ksh\ 450 = 450$) and irrational numbers (e.g., $ \sqrt{2} = \sqrt{2}$, $ 28\pi - 88 \approx 0.035$). It does NOT change whether a number is rational or irrational. The complete real number system classifies all numbers as either rational or irrational, and absolute value is a tool that works across the entire system to measure magnitude without regard to direction.	Some everyday numbers — such as the circumference of a sufuria with diameter 28 cm — cannot be written down completely because they are irrational: their decimal expansions neither terminate nor repeat. The real number system contains both rational numbers (expressible as fractions, with terminating or repeating decimals, like Ksh 60.50 or 34.5°C) and irrational numbers (like π and $\sqrt{2}$, which arise naturally from geometry and other mathematics). Mathematicians and Kenyan professionals — engineers, nurses, market traders, meteorologists — work accurately by using the best available rational approximation and by quantifying the approximation error using absolute value, so they always know exactly how close their working value is to the true value.

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